

TrES-3b

R-1876

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_140531 · created 2026-08-03T11:19:50+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456808.7312 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1454 +/- 0.009
Transit depth (Rp/Rs)^2	2.11 +/- 0.26 [%]
Orbital Inclination (inc)	84.19 +/- 0.87
Airmass coefficient 1 (a1)	1.054 +/- 0.014
Airmass coefficient 2 (a2)	-0.0386 +/- 0.0063
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#2 - [81, 82]
Optimal Method	PSF photometry
Transit Duration (day)	0.0692 +/- 0.0047

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1454 ± 0.009 vs 0.16309 (deviation 1.44σ)
Duration fit vs published	0.0692 d vs 0.0567 d (deviation 1.95σ)
Mid-time O-C	-3.6 min
Depth SNR	11.8
Residual scatter	1.3 %

Light curve

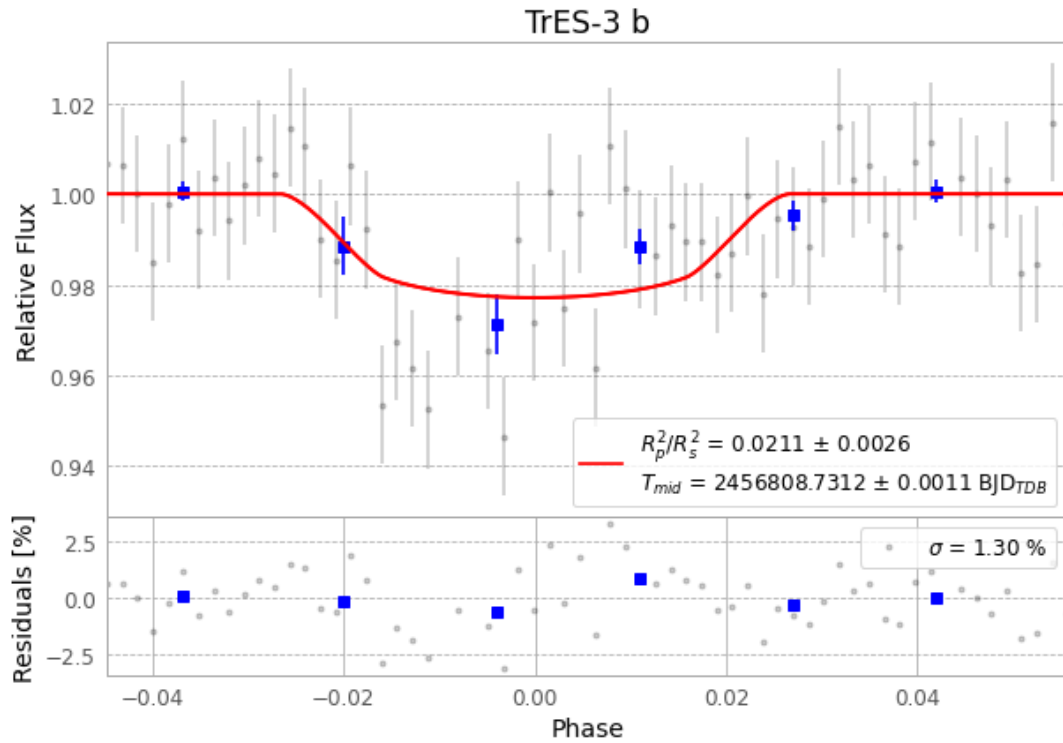


Figure 1 — FinalLightCurve_TrES-3 b_2014-05-30.png (SHA-256 0a35912b28c7b278...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [151, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 aa75e7b17cf39554...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

64 FITS frames (42 MB), TRES-3140531040312.FITS → TRES-3140531071214.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-140531.log (c18c252a65de...), FinalLightCurve_TrES-3 b_2014-05-30.png (0a35912b28c7...), FinalLightCurve_TrES-3 b_2014-05-30.csv (84b2dc348205...)

Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:19Z.